The Fourier transform of the density is the partial Fourier transform of the kernel

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Editorial note. The counted statement is the author's labelled lemma 'densityofoperator' (source lines 500-506); the proof environment begins on the next line after \end{lemma}, so the pairing is by adjacency.

Editorial note. The retained context is the author's earlier lemma identifying the Fourier representation of the Wigner transform, whose label is invoked at the end of the retained proof.

Editorial note. The retained preamble is the publisher layout adapter plus the native macro, operator and theorem block of the same pinned source.

Editorial note. Every label used inside the retained text is defined inside one of the retained spans, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: the partial Fourier representation of the density of an operator (author label densityofoperator).

Allowed standard prerequisites (editorial).

- Density of an operator. For an operator Q identified with its integral kernel Q(x,y), the density is rho_Q(x) := Q(x,x).. It is the object whose partial Fourier transforms are computed in the statement.
- Weighted Sobolev space H^0_{M,M}. H^0_{M,M} is the space of kernels whose weighted L2 norms up to order M are finite, with M = ceil((d+1)/2).. The second half of the proof extends the identity from Schwartz kernels to this space by density.
- Sobolev embedding via Cauchy-Schwarz. Since M > d/2, the sup over k of the integral in p of the weight is bounded by the weighted L2 norms of the kernel up to order M.. It is the estimate at source lines 515-525 that gives absolute convergence uniformly in k.

Lemma 0.1. *Suppose $Q \in \mathcal{H}^0_{M,M}(\mathbb{R}^d;\mathbb{C})$. Then the Wigner transform of Q satisfies, for all $k,\eta \in \mathbb{R}^d$,*

$$\left|\widehat{W[Q]}(k,\eta)\right| \lesssim_d \|Q\|_{\mathcal{H}^0_{M,M}}.$$

Lemma 0.2. *Suppose $Q \in \mathcal{H}^0_{M,M}(\mathbb{R}^d;\mathbb{C})$, where $M = \lceil (d+1)/2 \rceil$, identified with its integral kernel $Q = Q(x,y)$. Then the density $\rho_Q(x) := Q(x,x)$ satisfies, for all $k \in \mathbb{R}^d$,*

$$\widehat{\rho_Q}(k) = \frac{1}{(2\pi)^d} \int_p \widehat{Q}(k-p,p)\, dp = \frac{1}{(2\pi)^d} \int_p \widehat{Q}\left(\tfrac{k}{2}+p, \tfrac{k}{2}-p\right)\, dp,$$

both integrals converging absolutely.

Proof. First suppose that the kernel $Q(x,y)$ is a Schwartz function on $\mathbb{R}^d \times \mathbb{R}^d$. By Fourier inversion and Fubini's theorem,

$$\begin{aligned}\rho_Q(x) = Q(x,x) &= \frac{1}{(2\pi)^{2d}} \int_{k'} \int_p e^{i(k'+p)\cdot x} \widehat{Q}(k',p)\, dp\, dk' \\ &= \frac{1}{(2\pi)^d} \int_k e^{ik\cdot x} \left[\frac{1}{(2\pi)^d} \int_p \widehat{Q}(k-p,p)\, dp \right] dk,\end{aligned}$$

where the final equality follows from the substitution $k' = k - p$. Hence ρ_Q is the inverse Fourier transform of the bracketed function, which gives the first representation; the second follows from the change of variables $p \mapsto \frac{k}{2} - p$.

It remains to extend the identity to $Q \in \mathcal{H}^0_{M,M}$. Set $F_Q(k,p) := \widehat{Q}\left(\frac{k}{2}+p, \frac{k}{2}-p\right)$. Since $M > d/2$, the Cauchy–Schwarz inequality in p and the Sobolev embedding in k give

$$\begin{aligned}\sup_k \int_p |F_Q(k,p)|\, dp &\lesssim_d \hbar^{-d/2} \sup_k \left\| \langle \hbar p \rangle^M F_Q(k,\cdot) \right\|_{L^2_p} \\ &\lesssim_d \hbar^{-d/2} \sum_{|\beta| \le M} \left\| \partial_k^\beta \left(\langle \hbar p \rangle^M F_Q \right) \right\|_{L^2_{k,p}} \\ &\lesssim_d \|Q\|_{\mathcal{H}^0_{M,M}},\end{aligned}$$

where the final inequality bounds $\langle \hbar p \rangle^M$ by a finite sum of absolute monomials (M being an integer) and uses the relations

$$\widehat{\boldsymbol{x} Q}\left(\tfrac{k}{2}+p, \tfrac{k}{2}-p\right) = i \nabla_k F_Q(k,p), \qquad \widehat{\boldsymbol{\xi} Q}\left(\tfrac{k}{2}+p, \tfrac{k}{2}-p\right) = \hbar p\, F_Q(k,p),$$

iterated, together with Plancherel's theorem and the normalisation of the quantum Lebesgue norm. Thus the integrals in the statement converge absolutely, uniformly in k.

For each fixed k, the estimate above shows that the right-hand side of the stated identity depends continuously on $Q \in \mathcal{H}^0_{M,M}$, while, since $\widehat{\rho_Q}(k) = \widehat{W[Q]}(k,0)$ by the Fourier representation of the Wigner transform, Lemma 0.1 shows the same for the left-hand side. The identity therefore extends to all of $\mathcal{H}^0_{M,M}$ by density of Schwartz kernels. □

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